

SHARIF UNIVERSITY OF TECHNOLOGY

Department of Electrical Engineering

TECHNICAL REPORT

Single-Node Hadoop and Spark Deployment with Distributed Word Count

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ACADEMIC CONTEXT

Big Data Processing

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1 Processing model and Spark abstractions

Batch processing operates on a bounded dataset and suits historical analysis, ETL, and offline training. Streaming operates on an unbounded input and continuously produces updated results, often through micro-batches. Latency depends on ingestion, trigger interval, computation, and scheduling; calling a pipeline streaming does not establish a millisecond deadline. An RDD is an immutable partitioned distributed collection with lineage for recovery. Transformations construct a lazy computation graph; actions trigger execution. RDDs remain useful for custom low-level transformations and assignment constraints. They do not require the entire dataset to reside in RAM, and a lost partition can often be recomputed from its lineage. DataFrames expose named columns and a schema, allowing query optimization. Datasets add a typed interface in languages such as Scala and Java; Python does not have the same typed Dataset API. Type safety is not a guarantee against every runtime error, and optimization benefits depend on the operation used. The driver constructs the application, plans work, and coordinates jobs, stages, and tasks. The cluster manager allocates resources. Executors run tasks and may cache partitions. A shuffle redistributes data by key and can incur network transfer, serialization, disk I/O, and skew. reduceByKey can combine values locally before shuffle, whereas groupByKey usually transfers the raw grouped values.



Figure 1: Diagram or recorded result: Processing model and Spark abstractions.

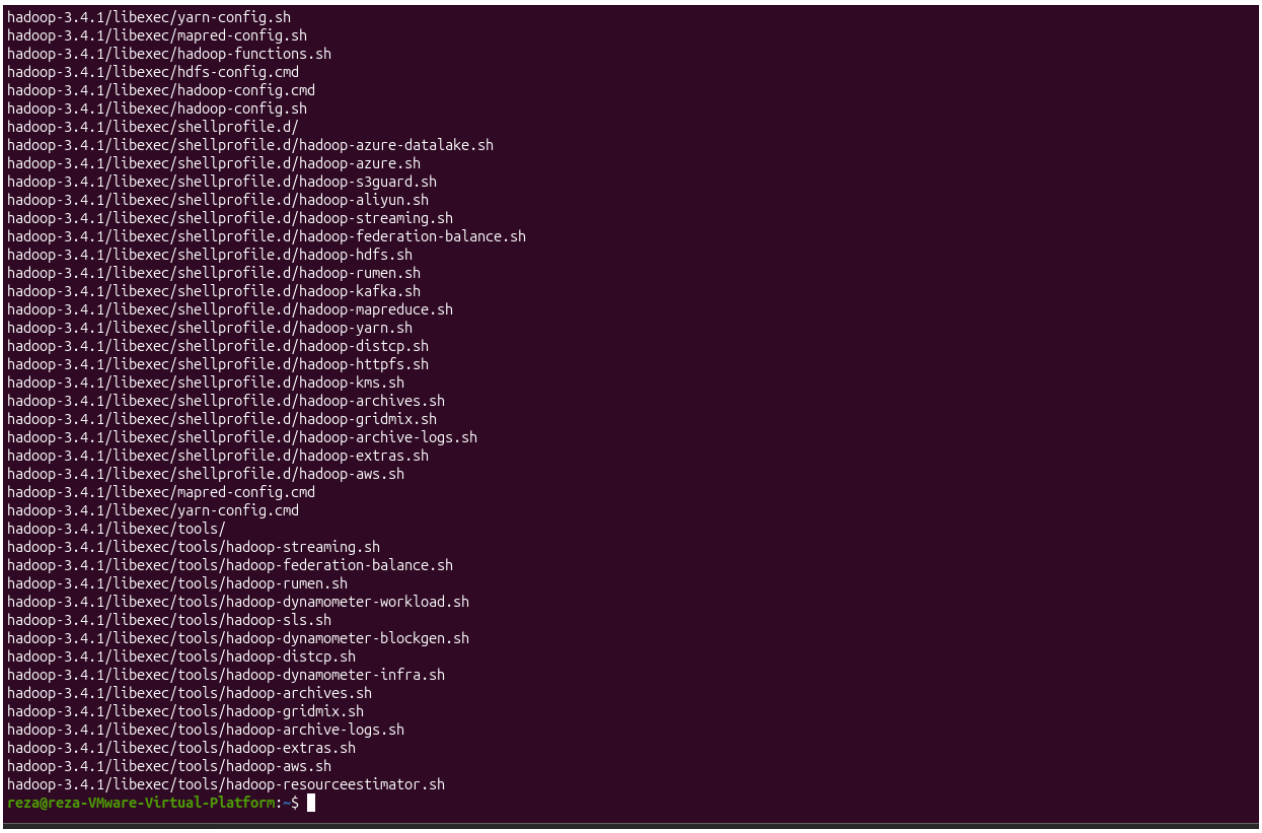


Figure 2: Diagram or recorded result: Processing model and Spark abstractions.

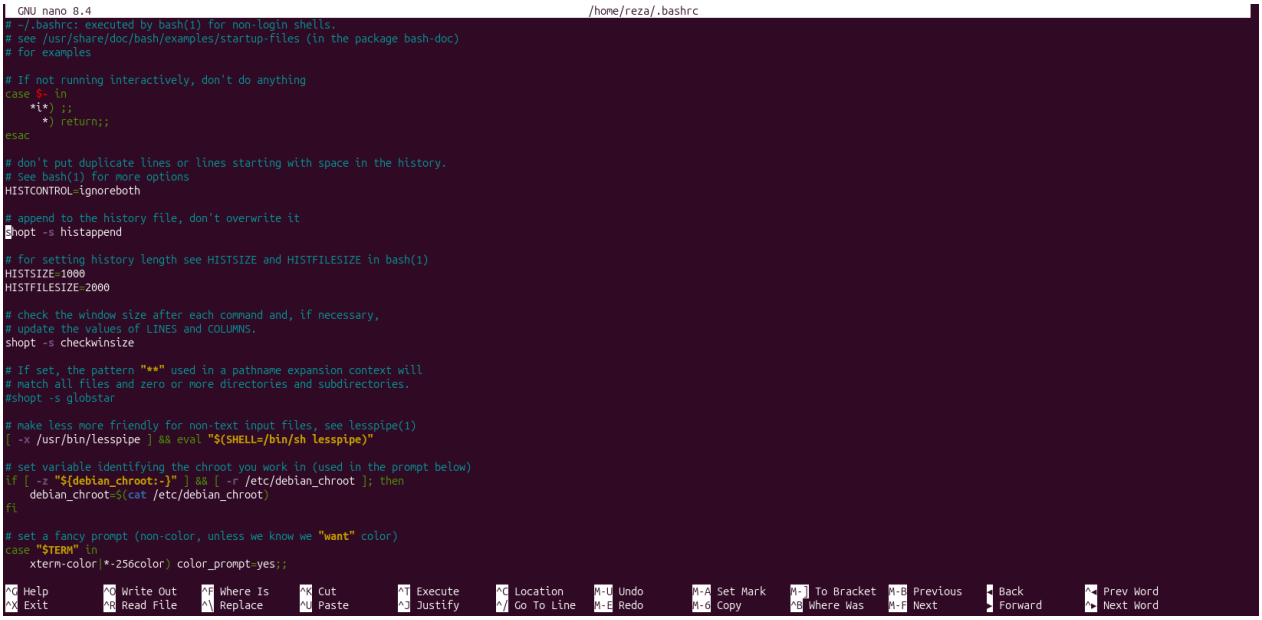


Figure 3: Diagram or recorded result: Processing model and Spark abstractions.

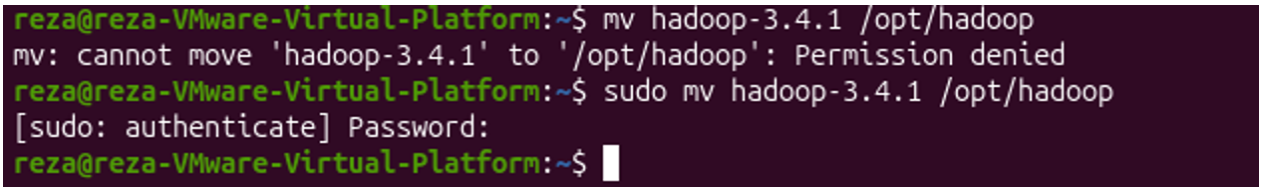


Figure 4: Implementation and output: Processing model and Spark abstractions.

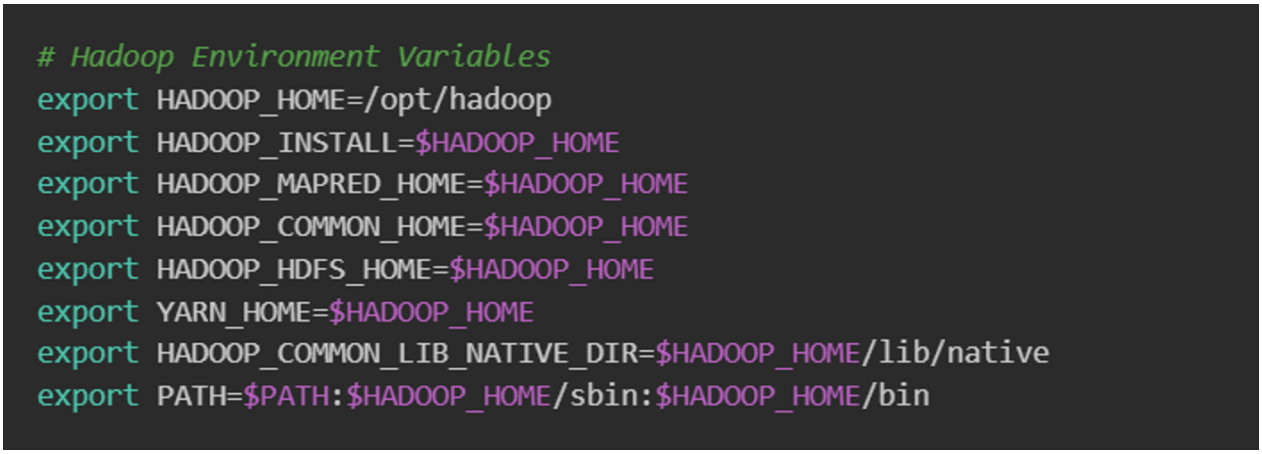


Figure 5: Diagram or recorded result: Processing model and Spark abstractions.

```
GNU nano 8.4                                    /home/reza/.bashrc
# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
#   sleep 10; alert
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo error)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&|]\s*alert$//'\'')"'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi


# Hadoop Environment Variables
export HADOOP_HOME=/opt/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
```

Figure 6: Diagram or recorded result: Processing model and Spark abstractions.

```
reza@reza-VMware-Virtual-Platform:~$ cd /opt/hadoop/etc/hadoop
reza@reza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$
```

Figure 7: Diagram or recorded result: Processing model and Spark abstractions.

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://<namenode-hostname-or-ip>:9000</value>
  </property>
</configuration>
```

Figure 8: Diagram or recorded result: Processing model and Spark abstractions.

```
GNU nano 8.4                                    core-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

Figure 9: Diagram or recorded result: Processing model and Spark abstractions.

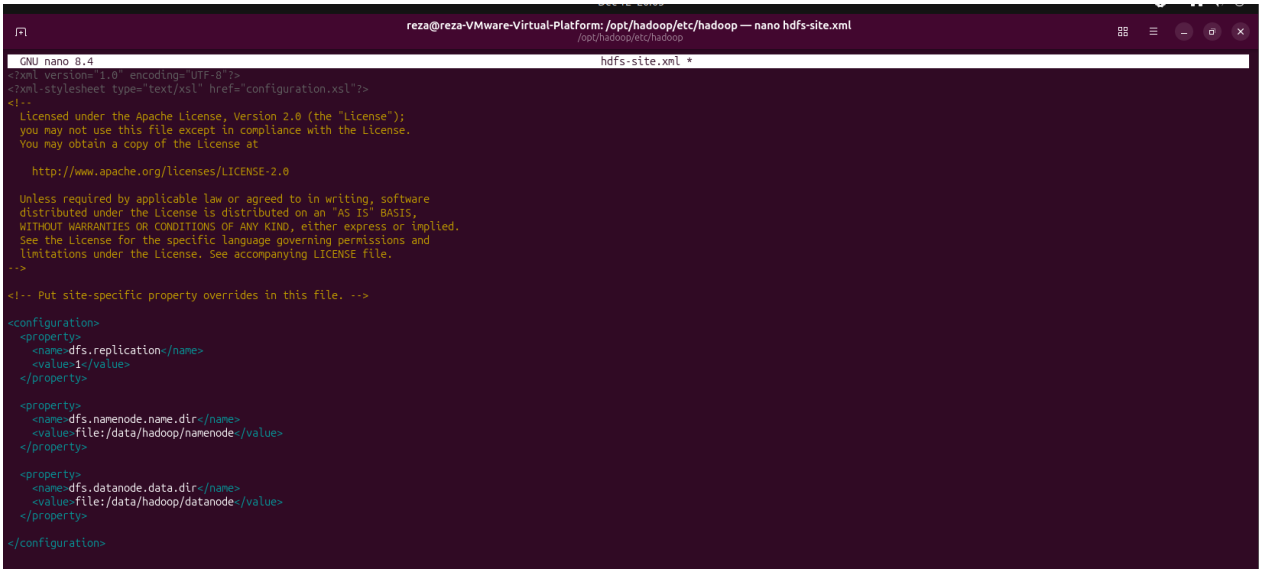


Figure 10: Diagram or recorded result: Processing model and Spark abstractions.

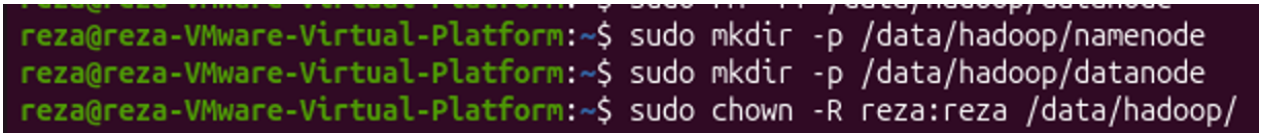


Figure 11: Implementation and output: Processing model and Spark abstractions.

2 Persistence, memory, and execution optimization

Caching is useful when a dataset is reused, especially in iterative processing. MEMORY_ONLY favors memory speed but may need recomputation when partitions do not fit. MEMORY_AND_DISK keeps overflow on disk. Serialized storage trades CPU overhead for a smaller representation in supported runtimes, while replicated levels retain additional copies. Python RDD objects are already serialized for storage; a serialized label does not mean a guaranteed compression ratio. RDD cache defaults and DataFrame defaults differ, so they should not be treated as one universal setting. Executor OOM may result from a large partition, skewed keys, excessive caching, oversized broadcasts, object overhead, or insufficient memory relative to concurrent tasks. Remedies include repartitioning, handling skew, reducing collected/broadcast data, unpersisting unused datasets, and adjusting resources. Increasing memory alone may leave the underlying skew unresolved. AQE uses runtime statistics to coalesce shuffle partitions, change suitable join strategies, and handle supported skewed joins. It does not optimize every operation automatically. Tungsten reduces overhead through compact representations and execution optimizations; whole-stage code generation can fuse supported operators into generated JVM code. These mechanisms do not mean that all computation becomes native machine code or that every garbage-collection cost disappears.

```
reza@reza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$ ls /usr/lib/jvm/
ls: cannot access '/usr/lib/jvm/': No such file or directory
```

Figure 12: Diagram or recorded result: Persistence, memory, and execution optimization.

```
reza@reza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$ sudo apt install openjdk-8-jdk -y
The following packages were automatically installed and are no longer required:
  linux-headers-6.17.0-5  linux-headers-6.17.0-5-generic  linux-modules-6.17.0-5-generic  linux-tools-6.17.0-5  linux-tools-6.17.0-5-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  openjdk-8-jdk

Installing dependencies:
  ca-certificates-java  java-common         libatk-wrapper-java-jni  libsm-dev   libxau-dev   libxdmcp-dev  openjdk-8-jdk-headless  openjdk-8-jre-headless  x11proto-dev       xtrans-dev
  fonts-dejavu-extra    libatk-wrapper-java  libice-dev               libx11-dev  libxcb1-dev  libxt-dev     openjdk-8-jre           uuid-dev                xorg-sgml-doctools

Suggested packages:
  default-jre  libsm-doc   libxcb-doc  openjdk-8-demo    visualvm     fonts-ipafont-gothic  fonts-wqy-microhei  fonts-indic
  libice-doc   libx11-doc  libxt-doc   openjdk-8-source  fonts-nanum  fonts-ipafont-mincho  fonts-wqy-zenhei

Summary:
  Upgrading: 0, Installing: 20, Removing: 0, Not Upgrading: 52
  Download size: 48.0 MB
  Space needed: 164 MB / 9,547 MB available

Get:1 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 ca-certificates-java all 20240118 [11.6 kB]
Get:2 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 fonts-dejavu-extra all 2.37-8 [1,947 kB]
Get:3 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 java-common all 0.76 [6,852 B]
Get:4 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libatk-wrapper-java all 0.42.1-2 [55.2 kB]
Get:5 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libatk-wrapper-java-jni amd64 0.42.1-2 [51.1 kB]
Get:6 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 xorg-sgml-doctools all 1:1.11-1.1 [10.9 kB]
Get:7 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 x11proto-dev all 2024.1-1 [606 kB]
Get:8 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libice-dev amd64 2:1.1.1-1 [54.0 kB]
Get:9 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 uuid-dev amd64 2.41-4ubuntu4 [56.3 kB]
Get:10 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libsm-dev amd64 2:1.2.6-1 [18.1 kB]
Get:11 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libxau-dev amd64 1:1.0.11-1build1 [10.0 kB]
Get:12 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libxdmcp-dev amd64 1:1.1.5-1build1 [26.9 kB]
Get:13 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 xtrans-dev all 1.6.0-1 [66.1 kB]
Get:14 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libxcb1-dev amd64 1.17.0-2build1 [91.8 kB]
```

Figure 13: Implementation and output: Persistence, memory, and execution optimization.

```
reza@reza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$ ls /usr/lib/jvm/
java-1.8.0-openjdk-amd64  java-8-openjdk-amd64
```

Figure 14: Diagram or recorded result: Persistence, memory, and execution optimization.

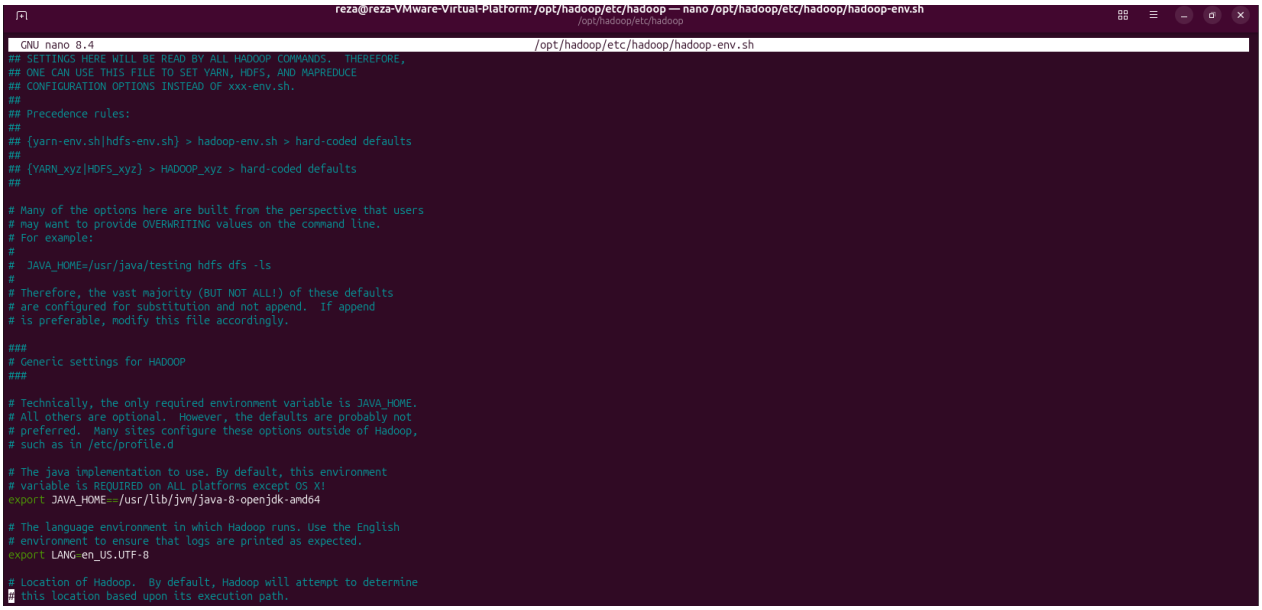


Figure 15: Diagram or recorded result: Persistence, memory, and execution optimization.

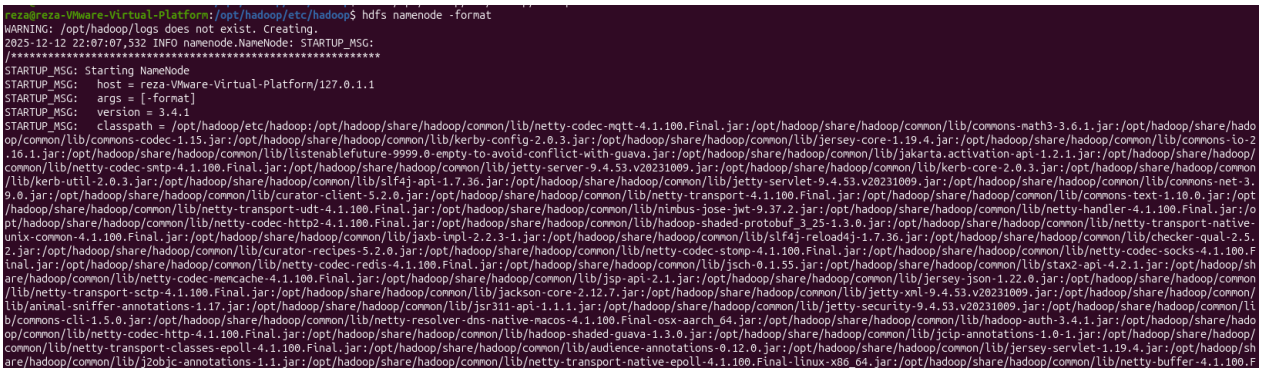


Figure 16: Diagram or recorded result: Persistence, memory, and execution optimization.



Figure 17: Diagram or recorded result: Persistence, memory, and execution optimization.

3 Deployment, file formats, and resource sharing

Standalone is suitable for a dedicated, relatively simple Spark cluster. YARN integrates with an existing Hadoop resource environment. Kubernetes offers container-oriented scheduling, but requires operational configuration for resources, storage, networking, and security. No deployment mode provides correct scaling or security merely by being selected. Parquet and ORC support column pruning, compression, and statistics-based data skipping; they are especially useful when reading selected columns from large analytical datasets. Delta Lake adds transactional table management and historical versions, subject to retained logs/data. Z-ordering can improve data locality and skipping; it is not an authorization mechanism. Warehouse versus lakehouse selection depends on governance, workload, infrastructure, and cost rather than a strict old-versus-new distinction. FIFO and Fair scheduling concern jobs sharing a Spark context; cluster-manager allocation between applications is a separate layer. Fair pools can specify weights and minimum shares. Tasks already running are normally not preempted simply because another pool submits work. Pool properties must be set in the appropriate submitting thread. Dynamic allocation requests or releases executors according to workload and configured limits; it requires a compatible way to preserve shuffle data and is not an instantaneous fairness guarantee.

```
reza@reza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$ start-dfs.sh
Starting namenodes on [localhost]
localhost: ssh: connect to host localhost port 22: Connection refused
Starting datanodes
localhost: ssh: connect to host localhost port 22: Connection refused
Starting secondary namenodes [reza-VMware-Virtual-Platform]
reza-VMware-Virtual-Platform: ssh: connect to host reza-vmware-virtual-platform port 22: Connection refused
2025-12-12 22:21:02,496 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Figure 18: Diagram or recorded result: Deployment, file formats, and resource sharing.

```
Files
eza-VMware-Virtual-Platform:/opt/hadoop/etc/hadoop$ sudo apt-get install openssh-server -y
 authenticate] Password:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.17.0-5 linux-headers-6.17.0-5-generic linux-modules-6.17.0-5-generic linux-tools-6.17.0-5 linux-tools-6.17.0-5-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 52 not upgraded.
Need to get 943 kB of archives.
After this operation, 7,527 kB of additional disk space will be used.
Get:1 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 openssh-sftp-server amd64 1:10.0p1-5ubuntu5 [37.5 kB]
Get:2 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 openssh-server amd64 1:10.0p1-5ubuntu5 [619 kB]
Get:3 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 ncurses-term all 6.5+20250216-2build1 [277 kB]
Get:4 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 ssh-import-id all 5.11-0ubuntu3 [10.1 kB]
Fetched 943 kB in 12s (81.8 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 185121 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a10.0p1-5ubuntu5_amd64.deb ...
Unpacking openssh-sftp-server (1:10.0p1-5ubuntu5) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a10.0p1-5ubuntu5_amd64.deb ...
Unpacking openssh-server (1:10.0p1-5ubuntu5) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.5+20250216-2build1_all.deb ...
Unpacking ncurses-term (6.5+20250216-2build1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu3_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu3) ...
Setting up openssh-sftp-server (1:10.0p1-5ubuntu5) ...
Setting up openssh-server (1:10.0p1-5ubuntu5) ...
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 981 and GID 65534.
Created symlink '/etc/systemd/system/sockets.target.wants/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
```

Figure 19: Implementation and output: Deployment, file formats, and resource sharing.

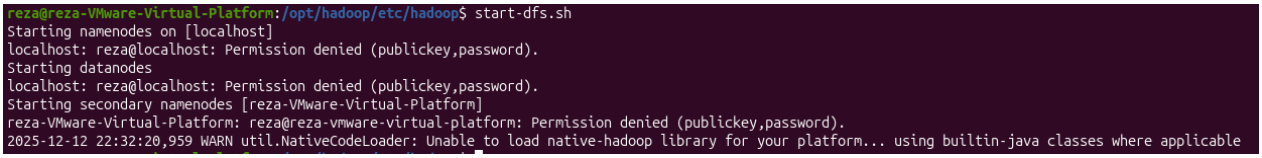


Figure 20: Diagram or recorded result: Deployment, file formats, and resource sharing.

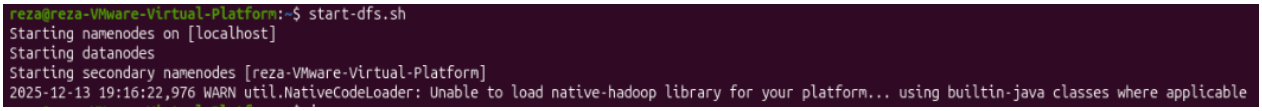


Figure 21: Diagram or recorded result: Deployment, file formats, and resource sharing.

4 Part A: installing and configuring Hadoop

The supplied installation record follows hadoop_installation.md. Hadoop 3.4.1 was downloaded, extracted, and placed under /opt/hadoop. Environment variables were added to ~/.bashrc. Linux paths are case-sensitive. Use the actual /opt/hadoop and /data/hadoop directories consistently in commands and configuration. Moving files under /opt and configuring protected directories requires appropriate ownership or sudo. The core configuration sets fs.defaultFS to hdfs://localhost:9000. The HDFS configuration uses replication=1 for the single-machine laboratory and separate NameNode/DataNode directories under /data/hadoop. Directory ownership must permit the Hadoop account to access them. JAVA_HOME was configured for the Java installation used by Hadoop. The source describes Java 8 for this stage and Java 11 for a later Spark stage; each component's environment must be checked explicitly rather than assuming one shell value works for all services. After configuring the worker as localhost, the new NameNode was formatted once. The first start-dfs.sh attempt failed because an SSH service was absent. Installing openssh-server addressed that problem. A later passwordless-SSH failure was addressed by generating a local key and adding its public key to authorized_keys. Existing keys should be retained rather than overwritten when adapting the procedure. Once SSH access was configured, start-dfs.sh launched the services and hdfs dfsadmin -report displayed the DataNode status and capacity.



Figure 22: Diagram or recorded result: Part A: installing and configuring Hadoop.

```
-------------------------------------------------
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: reza-VMware-Virtual-Platform
Decommission Status : Normal
Configured Capacity: 26238750720 (24.44 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 15185854464 (14.14 GB)
DFS Remaining: 9694076928 (9.03 GB)
DFS Used%: 0.00%
DFS Remaining%: 36.95%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sat Dec 13 19:20:13 IRST 2025
Last Block Report: Sat Dec 13 19:16:20 IRST 2025
Num of Blocks: 0
```

Figure 23: Diagram or recorded result: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ hdfs dfs -mkdir -p /user/reza/lab/
2025-12-13 19:49:47,247 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Figure 24: Diagram or recorded result: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ hdfs dfs -ls /user/reza/lab/
2025-12-13 19:54:22,339 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 reza supergroup         35 2025-12-13 19:51 /user/reza/lab/sample.txt
```

Figure 25: Diagram or recorded result: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ sudo apt install openjdk-11-jdk -y
Installing:
  openjdk-11-jdk

Installing dependencies:
  openjdk-11-jdk-headless  openjdk-11-jre  openjdk-11-jre-headless

Suggested packages:
  openjdk-11-demo  openjdk-11-source  visualvm  fonts-ipafont-gothic  fonts-ipafont-mincho  fonts-wqy-microhei  | fonts-wqy-zenhei  fonts-indic

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 52
  Download size: 119 MB
  Space needed: 263 MB / 9,691 MB available

Get:1 http://ir.archive.ubuntu.com/ubuntu questing-updates/universe amd64 openjdk-11-jre-headless amd64 11.0.29+7-1ubuntu1~25.10 [42.4 MB]
Get:2 http://ir.archive.ubuntu.com/ubuntu questing-updates/universe amd64 openjdk-11-jre amd64 11.0.29+7-1ubuntu1~25.10 [212 kB]
Get:3 http://ir.archive.ubuntu.com/ubuntu questing-updates/universe amd64 openjdk-11-jdk-headless amd64 11.0.29+7-1ubuntu1~25.10 [73.9 MB]
Get:4 http://ir.archive.ubuntu.com/ubuntu questing-updates/universe amd64 openjdk-11-jdk amd64 11.0.29+7-1ubuntu1~25.10 [2,980 kB]
Fetched 119 MB in 1min 13s (1,626 kB/s)
Selecting previously unselected package openjdk-11-jre-headless:amd64.
(Reading database ... 148010 files and directories currently installed.)
Preparing to unpack .../openjdk-11-jre-headless_11.0.29+7-1ubuntu1~25.10_amd64.deb ...
Unpacking openjdk-11-jre-headless:amd64 (11.0.29+7-1ubuntu1~25.10) ...
Selecting previously unselected package openjdk-11-jre:amd64.
Preparing to unpack .../openjdk-11-jre_11.0.29+7-1ubuntu1~25.10_amd64.deb ...
Unpacking openjdk-11-jre:amd64 (11.0.29+7-1ubuntu1~25.10) ...
Selecting previously unselected package openjdk-11-jdk-headless:amd64.
Preparing to unpack .../openjdk-11-jdk-headless_11.0.29+7-1ubuntu1~25.10_amd64.deb ...
Unpacking openjdk-11-jdk-headless:amd64 (11.0.29+7-1ubuntu1~25.10) ...
Selecting previously unselected package openjdk-11-jdk:amd64.
```

Figure 26: Implementation and output: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ wget https://dlcdn.apache.org/spark/spark-3.5.4/spark-3.5.4-bin-hadoop3.tgz
--2025-12-13 20:20:03--  https://dlcdn.apache.org/spark/spark-3.5.4/spark-3.5.4-bin-hadoop3.tgz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-13 20:20:03 ERROR 404: Not Found.
```

Figure 27: Implementation and output: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ wget https://dlcdn.apache.org/spark/spark-3.5.7/spark-3.5.7-bin-hadoop3.tgz
--2025-12-13 20:52:04--  https://dlcdn.apache.org/spark/spark-3.5.7/spark-3.5.7-bin-hadoop3.tgz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 400914067 (382M) [application/x-gzip]
Saving to: 'spark-3.5.7-bin-hadoop3.tgz.3'

spark-3.5.7-bin-hadoop3.tgz.3                  100%[=========================================================================================================>] 382.34M   790KB/s    in 15m 54s

2025-12-13 21:08:01 (410 KB/s) - 'spark-3.5.7-bin-hadoop3.tgz.3' saved [400914067/400914067]
```

Figure 28: Implementation and output: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ tar xvf spark-3.5.7-bin-hadoop3.tgz
spark-3.5.7-bin-hadoop3/
spark-3.5.7-bin-hadoop3/R/
spark-3.5.7-bin-hadoop3/R/lib/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/R/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/R/SparkR
spark-3.5.7-bin-hadoop3/R/lib/SparkR/R/SparkR.rdb
spark-3.5.7-bin-hadoop3/R/lib/SparkR/R/SparkR.rdx
spark-3.5.7-bin-hadoop3/R/lib/SparkR/profile/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/profile/general.R
spark-3.5.7-bin-hadoop3/R/lib/SparkR/profile/shell.R
spark-3.5.7-bin-hadoop3/R/lib/SparkR/tests/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/tests/testthat/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/tests/testthat/test_basic.R
spark-3.5.7-bin-hadoop3/R/lib/SparkR/DESCRIPTION
spark-3.5.7-bin-hadoop3/R/lib/SparkR/worker/
spark-3.5.7-bin-hadoop3/R/lib/SparkR/worker/daemon.R
```

Figure 29: Implementation and output: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ mv spark-3.5.7-bin-hadoop3 /opt/spark
mv: cannot move 'spark-3.5.7-bin-hadoop3' to '/opt/spark': Permission denied
reza@reza-VMware-Virtual-Platform:~$ sudo mv spark-3.5.7-bin-hadoop3 /opt/spark
Show Apps nticate] Password:
Mware-Virtual-Platform:~$
```

Figure 30: Implementation and output: Part A: installing and configuring Hadoop.

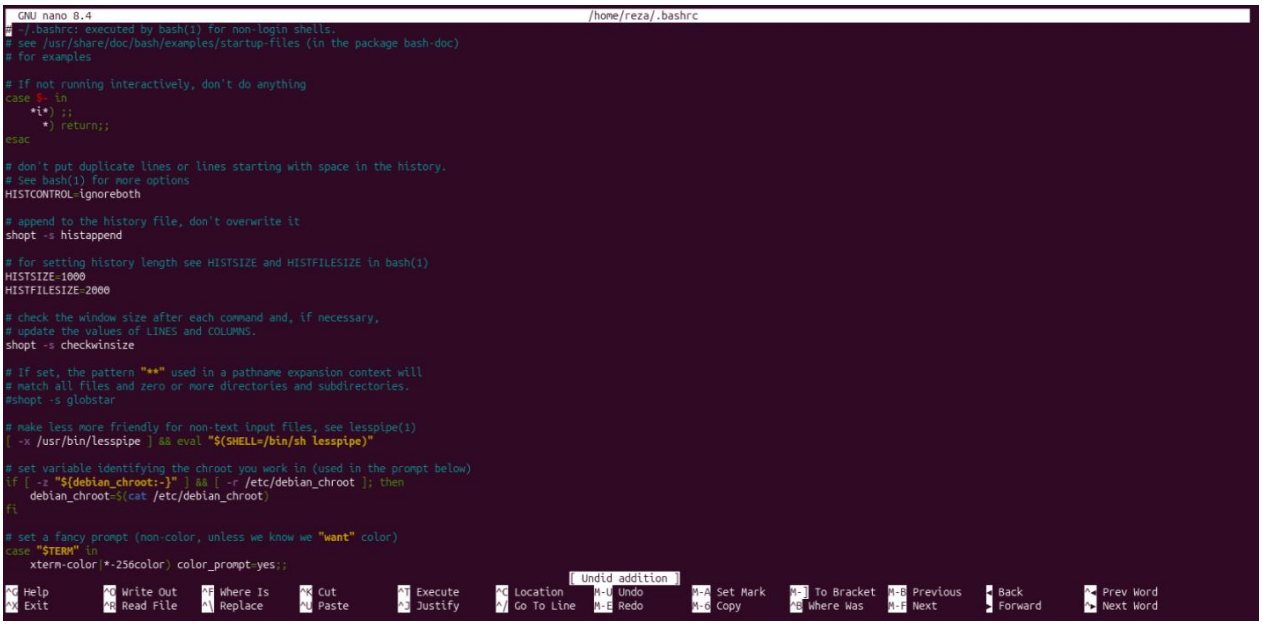


Figure 31: Diagram or recorded result: Part A: installing and configuring Hadoop.

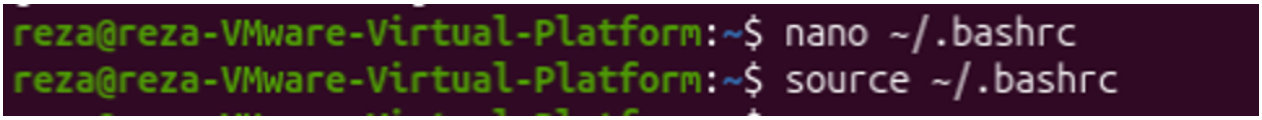


Figure 32: Diagram or recorded result: Part A: installing and configuring Hadoop.



Figure 33: Implementation and output: Part A: installing and configuring Hadoop.

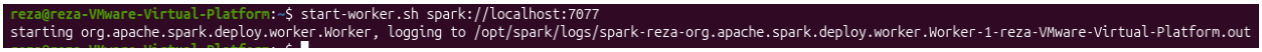


Figure 34: Implementation and output: Part A: installing and configuring Hadoop.

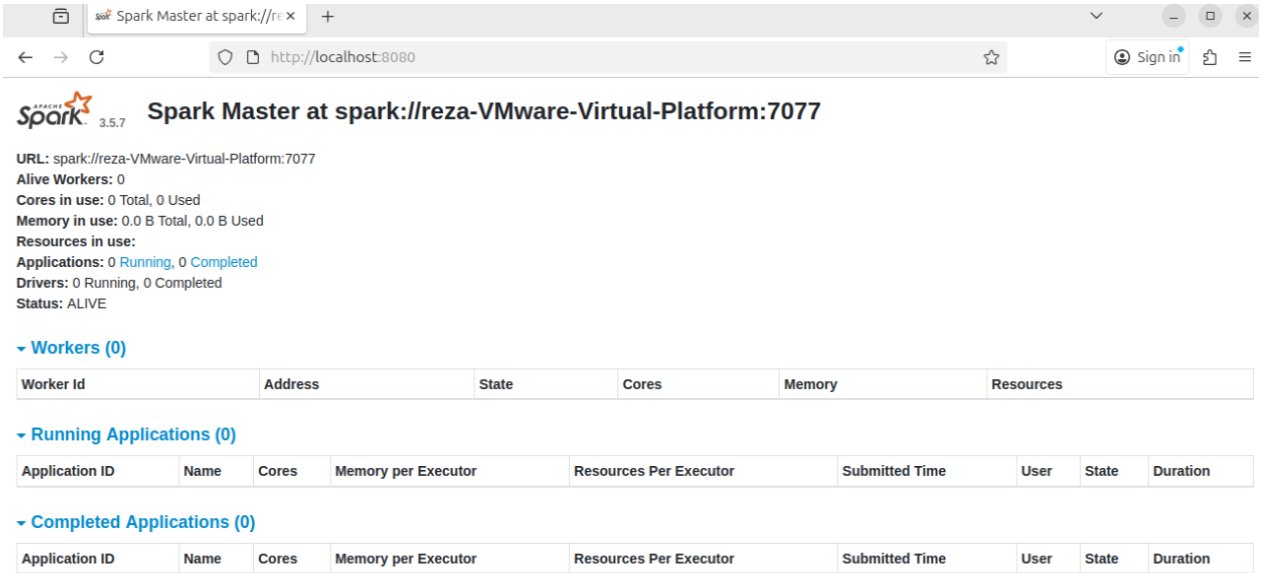


Figure 35: Implementation and output: Part A: installing and configuring Hadoop.

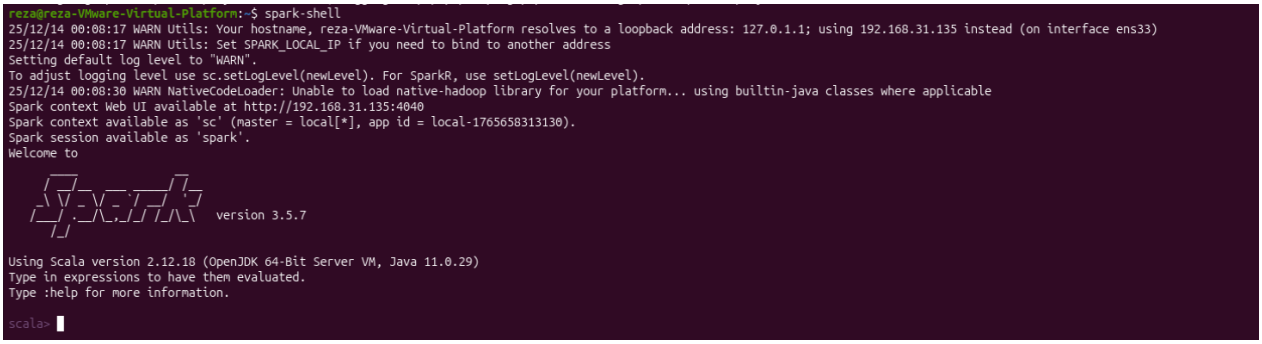


Figure 36: Implementation and output: Part A: installing and configuring Hadoop.

```
scala> val data = sc.parallelize(Seq("Hello", "World", "Hello", "Spark"))
data: org.apache.spark.rdd.RDD[String] = ParallelCollectionRDD[0] at parallelize at <console>:23

scala> val counts = data.map(word => (word, 1)).reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[2] at reduceByKey at <console>:23

scala> counts.collect().foreach(println)
(Hello,2)
(World,1)
(Spark,1)
```

Figure 37: Implementation and output: Part A: installing and configuring Hadoop.

```
reza@reza-VMware-Virtual-Platform:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [reza-VMware-Virtual-Platform]
2025-12-14 00:23:47,697 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Figure 38: Diagram or recorded result: Part A: installing and configuring Hadoop.

```
scala> :paste
// Entering paste mode (ctrl-D to finish)

val rdd = sc.textFile("hdfs://localhost:9000/user/reza/lab/sample.txt")

val counts = rdd
  .flatMap(_.split("\\s+"))
  .filter(_.nonEmpty)
  .map(w => (w.toLowerCase, 1))
  .reduceByKey(_ + _)
  .sortBy({ case (_, c) => -c })

counts.collect().foreach(println)

// Exiting paste mode, now interpreting.

(spark,3)
(hello,2)
(hdfs,1)
rdd: org.apache.spark.rdd.RDD[String] = hdfs://localhost:9000/user/reza/lab/sample.txt MapPartitionsRDD[6] at textFile at <pastie>:23
counts: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[15] at sortBy at <pastie>:30
```

Figure 39: Implementation and output: Part A: installing and configuring Hadoop.

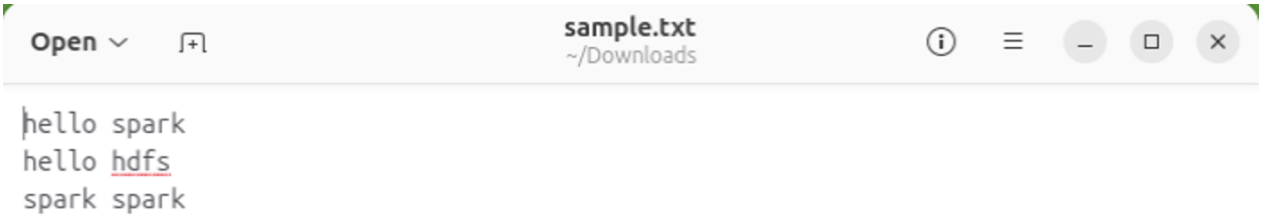


Figure 40: Implementation and output: Part A: installing and configuring Hadoop.

5 HDFS upload test

The report creates sample.txt containing three lines, creates /user/reza/lab/, uploads the file, and lists the directory. The listing confirms that sample.txt appears in HDFS. This is a functional upload/listing test; it does not establish replication across machines because the documented topology has one DataNode.

```
reza@reza-VMware-Virtual-Platform:~$ sudo apt install pip
[sudo: authenticate] Password:
Note, selecting 'python3-pip' instead of 'pip'
Installing:
  python3-pip

Installing dependencies:
  build-essential          g++-x86-64-linux-gnu     libalgorithm-diff-xs-perl  libfile-fcntllock-perl  liblsan0           lto-disabled-list
  bzip2                    gcc                      libalgorithm-merge-perl    libgcc-15-dev           libpython3-dev     make
  dpkg-dev                 gcc-15                   libasan8                   libhwasan0              libpython3.13-dev  python3-dev
  fakeroot                 gcc-15-x86-64-linux-gnu  libcc1-0                   libitm1                 libquadmath0       python3-packaging
  g++                      gcc-x86-64-linux-gnu     libdpkg-perl               libjs-jquery            libstdc++-15-dev   python3-wheel
  g++-15                   javascript-common        libexpat1-dev              libjs-sphinxdoc         libtsan2           python3.13-dev
  g++-15-x86-64-linux-gnu  libalgorithm-diff-perl   libfakeroot                libjs-underscore        libubsan1          zlib1g-dev

Suggested packages:
  bzip2-doc                g++-multilib     gcc-multilib  libtool  gcc-doc          gdb-x86-64-linux-gnu  | httpd  libstdc++-15-doc
  debian-keyring           g++-15-multilib  autoconf      flex     gcc-15-multilib  apache2               git     make-doc
  debian-tag2upload-keyring  gcc-15-doc     automake      bison    gcc-15-locales   | lighttpd            bzr     python3-setuptools

Summary:
  Upgrading: 0, Installing: 43, Removing: 0, Not Upgrading: 52
  Download size: 68.2 MB
  Space needed: 245 MB / 7,726 MB available

Continue? [Y/n] y
Get:1 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libcc1-0 amd64 15.2.0-4ubuntu4 [47.4 kB]
Get:2 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libitm1 amd64 15.2.0-4ubuntu4 [29.3 kB]
Get:3 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libasan8 amd64 15.2.0-4ubuntu4 [3,071 kB]
Get:4 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 liblsan0 amd64 15.2.0-4ubuntu4 [1,360 kB]
Get:5 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libtsan2 amd64 15.2.0-4ubuntu4 [2,757 kB]
Get:6 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libubsan1 amd64 15.2.0-4ubuntu4 [1,210 kB]
Get:7 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libhwasan0 amd64 15.2.0-4ubuntu4 [1,685 kB]
Get:8 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libquadmath0 amd64 15.2.0-4ubuntu4 [153 kB]
Get:9 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 libgcc-15-dev amd64 15.2.0-4ubuntu4 [2,864 kB]
Get:10 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 gcc-15-x86-64-linux-gnu amd64 15.2.0-4ubuntu4 [25.4 MB]
Get:11 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 gcc-15 amd64 15.2.0-4ubuntu4 [521 kB]
Get:12 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 gcc-x86-64-linux-gnu amd64 4:15.2.0-4ubuntu1 [1,208 B]
Get:13 http://ir.archive.ubuntu.com/ubuntu questing/main amd64 gcc amd64 4:15.2.0-4ubuntu1 [5,024 B]
```

Figure 41: Implementation and output: HDFS upload test.

```
reza@reza-VMware-Virtual-Platform:~$ pip install pyspark
error: externally-managed-environment

× This environment is externally managed
╰─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Debian-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have python3-full installed.

    If you wish to install a non-Debian packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.

    See /usr/share/doc/python3.13/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of b
aking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.
```

Figure 42: Implementation and output: HDFS upload test.

6 Part B: Spark installation and shell tests

The Spark guide initially referred to version 3.5.4. A download attempt failed, and the report switched to Spark 3.5.7. All subsequent extraction paths and example JAR names must therefore use 3.5.7 consistently. SPARK_HOME and PATH were added to ~/.bashrc. start-master.sh launched the master and start-worker.sh spark://localhost:7077 registered the worker. The browser interface on port 8080 showed the deployment. A Scala shell counted the list Hello, World, Hello, Spark. A further Scala test loaded sample.txt from HDFS and applied flatMap, filter, map, reduceByKey, and sorting. The correct sample counts are spark=3, hello=2, hdfs=1.

```
reza@reza-VMware-Virtual-Platform:~$ pip install pyspark --break-system-packages
Defaulting to user installation because normal site-packages is not writeable
Collecting pyspark
  Downloading pyspark-4.0.1.tar.gz (434.2 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 434.2/434.2 MB 1.1 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... error
  error: subprocess-exited-with-error

  × python setup.py egg_info did not run successfully.
  │ exit code: 1
  ╰─> [15 lines of output]
      Traceback (most recent call last):
        File "<string>", line 2, in <module>
          exec(compile('''
          ~~~~^^^^^^^^^^^^
          # This is <pip-setuptools-caller> -- a caller that pip uses to run setup.py
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
          ...<32 lines>...
          exec(compile(setup_py_code, filename, "exec"))
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
          ''' % ('/tmp/pip-install-xj69d6ei/pyspark_db21e661173b4d3090942b802f9ed63a/setup.py',), "<pip-setuptools-caller>", "exec"))
          ^^^^^^^^^^^^^^^^^^^
```

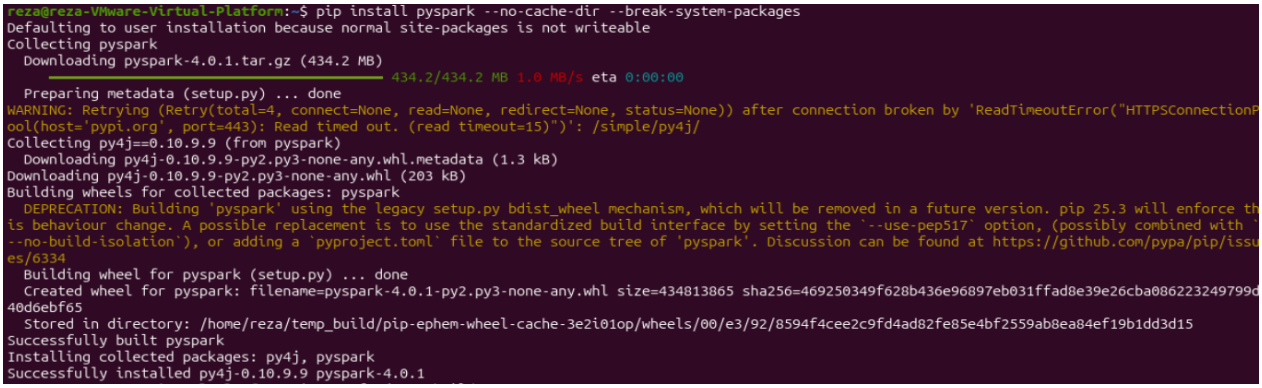


Figure 45: Implementation and output: Part B: Spark installation and shell tests.

```
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi


# Hadoop Environment Variables
export HADOOP_HOME=/opt/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin

# Spark Environment Variables
export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

export PYSPARK_PYTHON=/usr/bin/python3
```

Figure 46: Implementation and output: Part B: Spark installation and shell tests.

7 Python environment troubleshooting

The initial global pip installation was refused because the operating system manages that Python environment. The submission used –break-system-packages after installing setuptools. For a reproducible setup, use an isolated environment and a matching PySpark version instead of modifying the managed interpreter:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -upgrade pip setuptools
python -m pip install pyspark==3.5.7
```

The reported PySpark shell test parallelizes [1,2,3,4,5], squares each entry, and prints [1,4,9,16,25]. That confirms the basic Python-to-Spark computation for the tested environment.

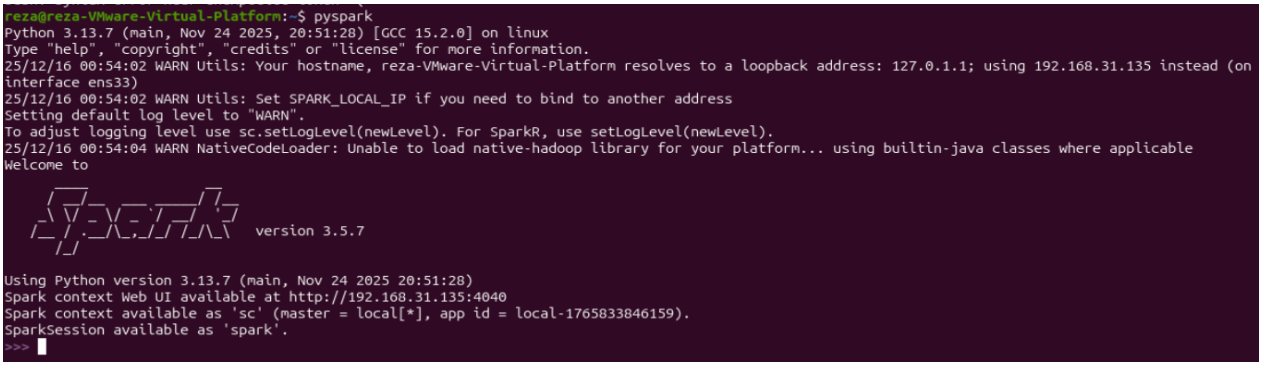


Figure 47: Implementation and output: Python environment troubleshooting.

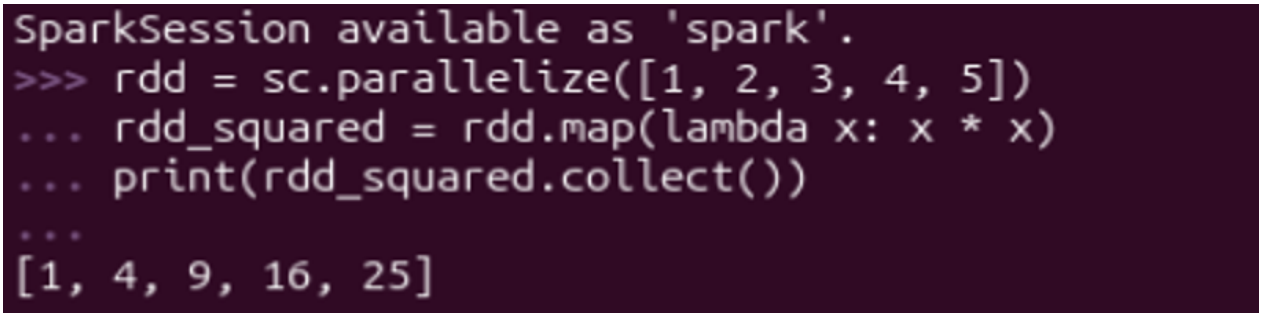


Figure 48: Implementation and output: Python environment troubleshooting.

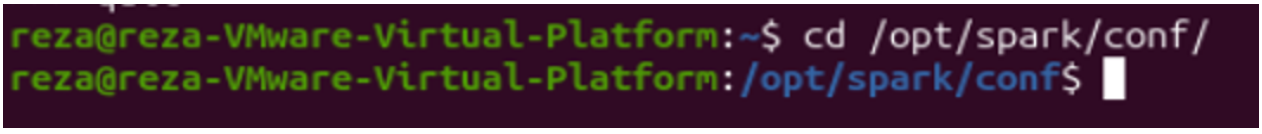


Figure 49: Implementation and output: Python environment troubleshooting.

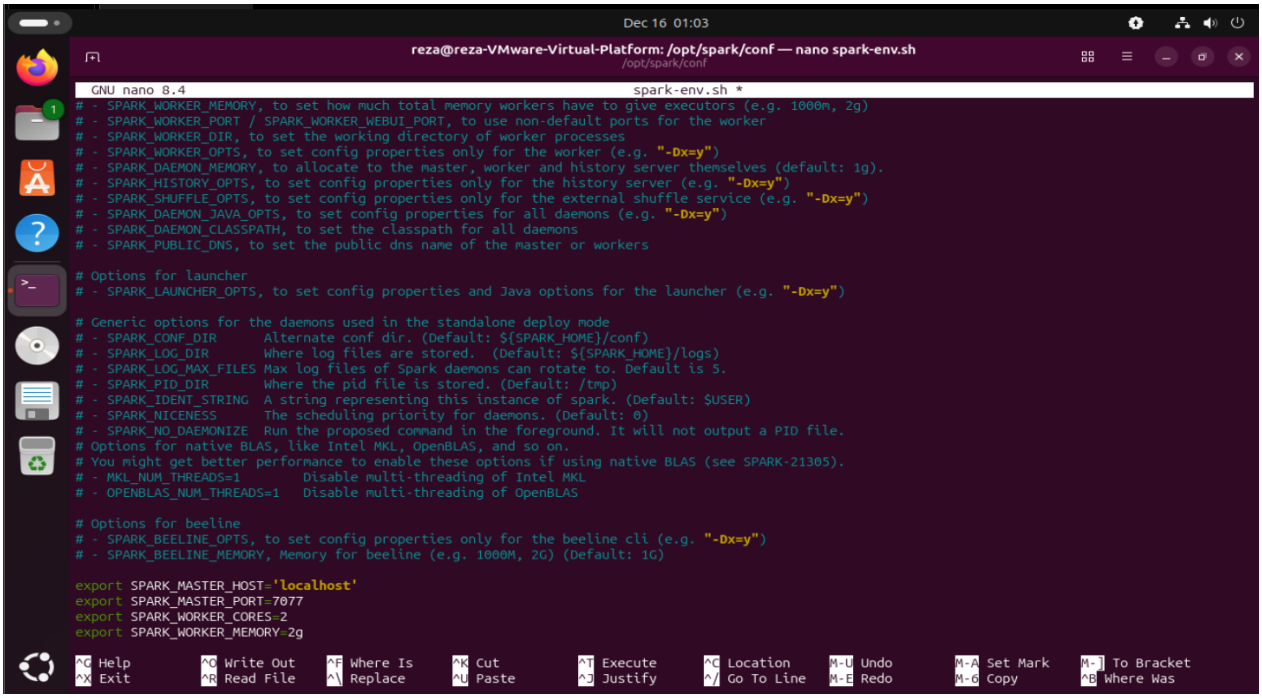


Figure 50: Implementation and output: Python environment troubleshooting.

8 Master/worker configuration and final integration

spark-env.sh and workers were created from their templates. The workers file contains localhost because the master and worker share one machine. A hosts entry maps master to the local address. The master UI on port 8080, worker UI on port 8081, and the master's JSON endpoint were used for inspection. spark-defaults.conf sets spark.master to spark://master:7077 and a local driver address. SPARK_LOCAL_IP=127.0.0.1 is appropriate only to this single-host example; a multi-host deployment needs reachable addresses. The final SparkPi example requests 1 GB for the driver and executor and uses 10 samples/partitions as stated by the example invocation. The source reports Pi is roughly 3.1406631406631407. The consistent command for the installed version is:

```
spark-submit -class org.apache.spark.examples.SparkPi \
-master spark://master:7077 -executor-memory 1G \
-driver-memory 1G \
/opt/spark/examples/jars/spark-examples_2.12-3.5.7.jar 10
```

The installation evidence supports a working classroom HDFS/Spark integration on the documented host. It does not establish multi-node throughput, fault tolerance, authentication, or production readiness. All deployment screenshots and the full troubleshooting sequence are retained.

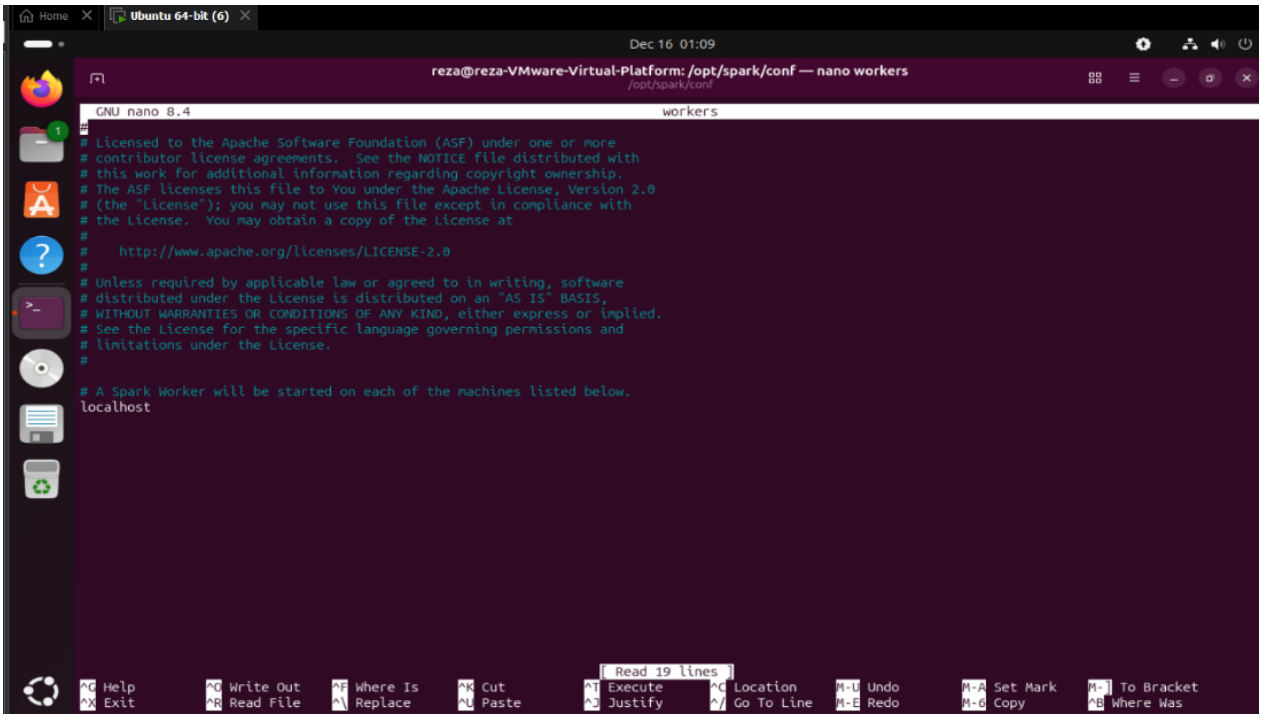


Figure 51: Implementation and output: Master/worker configuration and final integration.

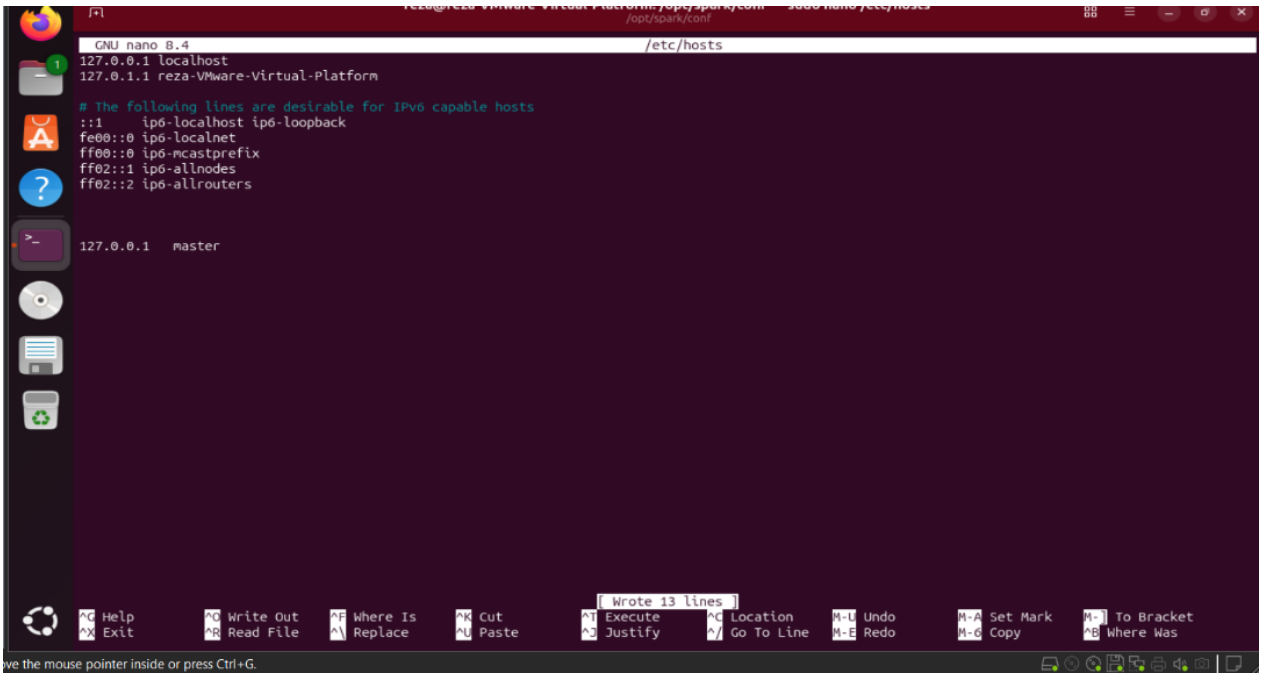


Figure 52: Implementation and output: Master/worker configuration and final integration.



Figure 53: Implementation and output: Master/worker configuration and final integration.



Figure 54: Implementation and output: Master/worker configuration and final integration.

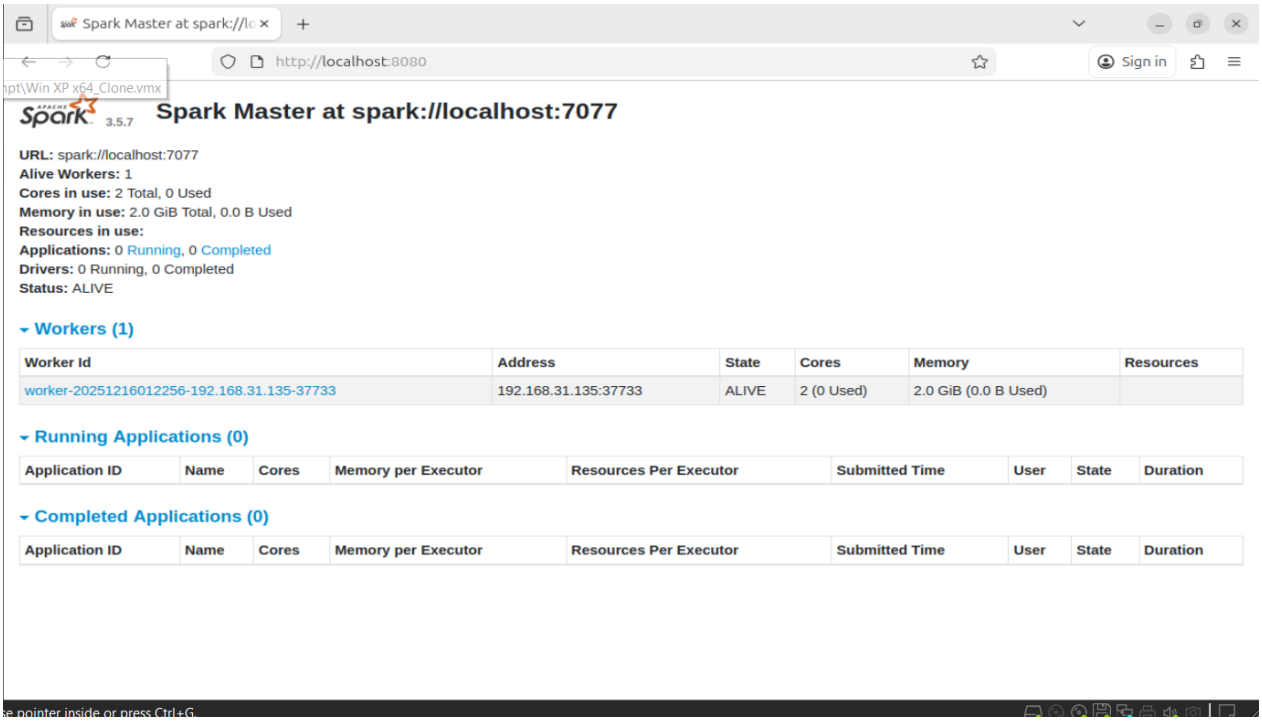


Figure 55: Implementation and output: Master/worker configuration and final integration.



Figure 56: Implementation and output: Master/worker configuration and final integration.

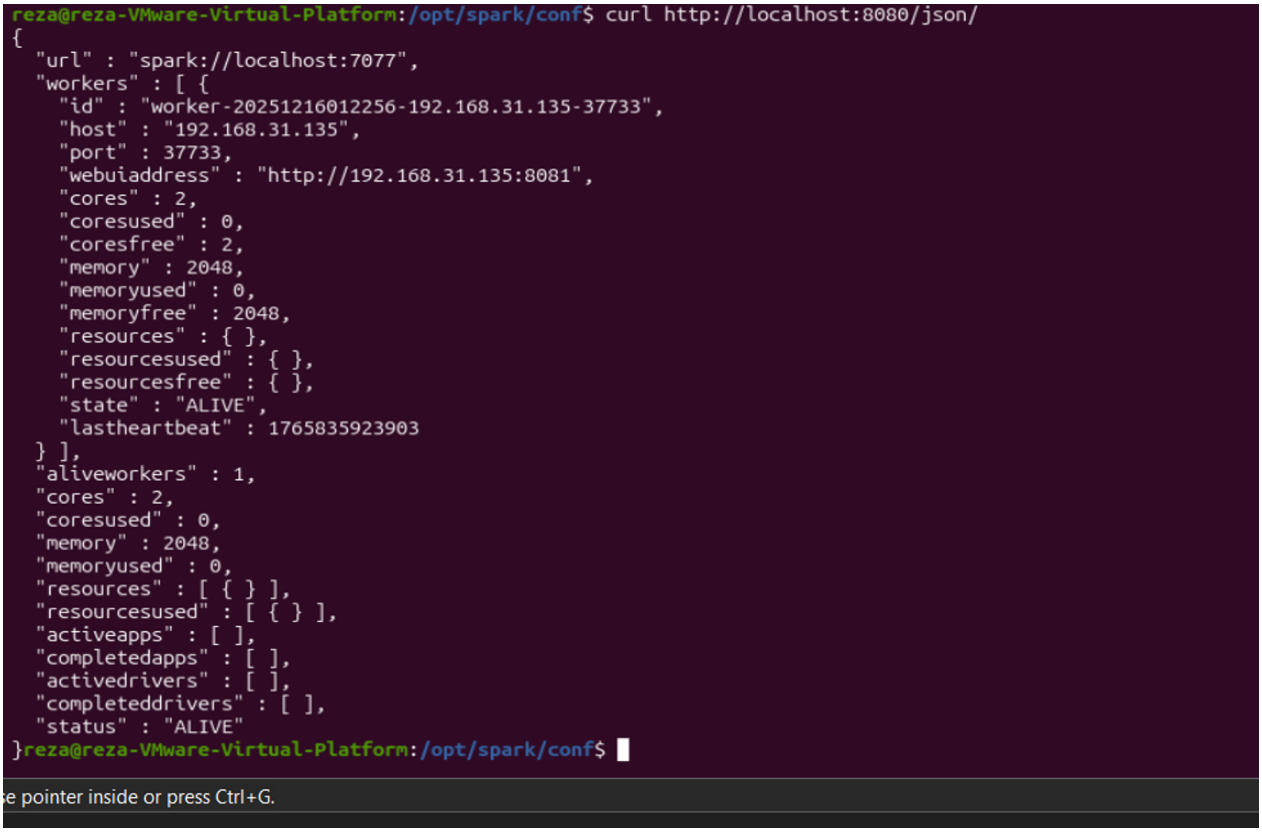


Figure 57: Implementation and output: Master/worker configuration and final integration.

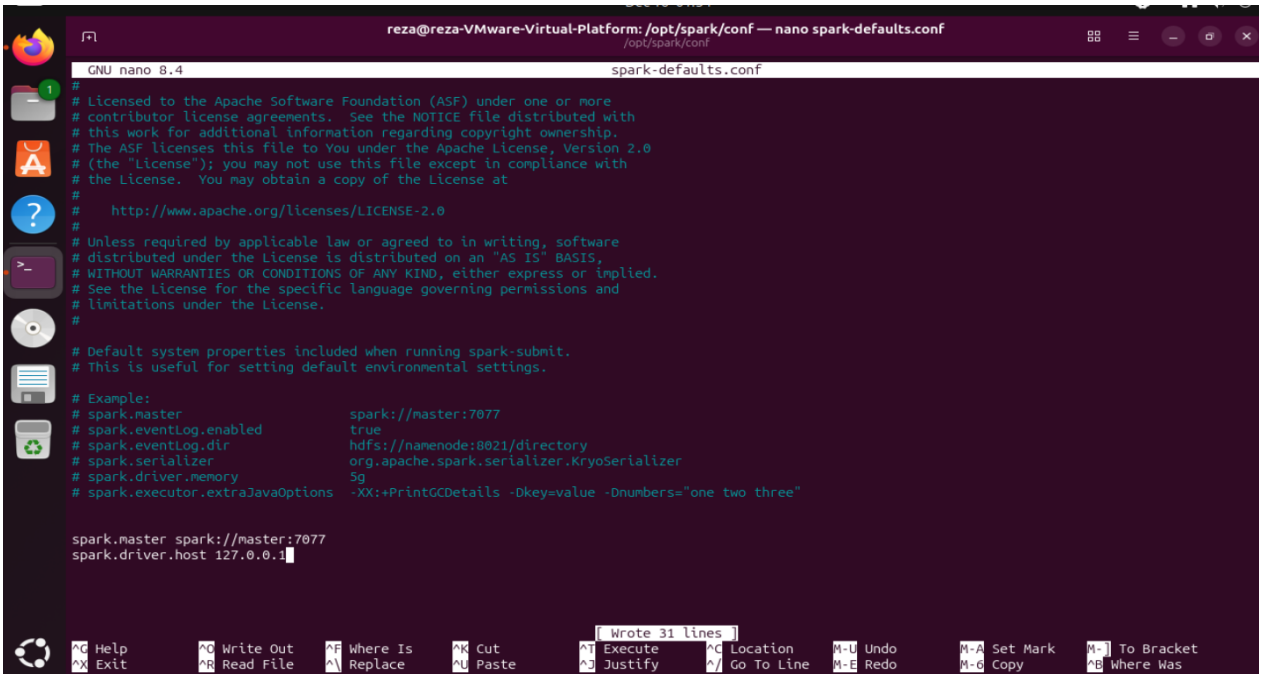


Figure 58: Implementation and output: Master/worker configuration and final integration.



Figure 59: Implementation and output: Master/worker configuration and final integration.

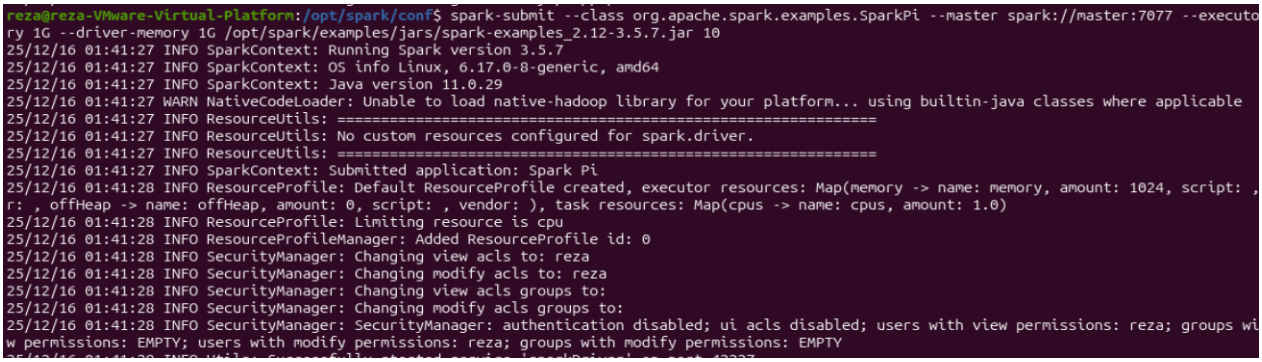


Figure 60: Implementation and output: Master/worker configuration and final integration.

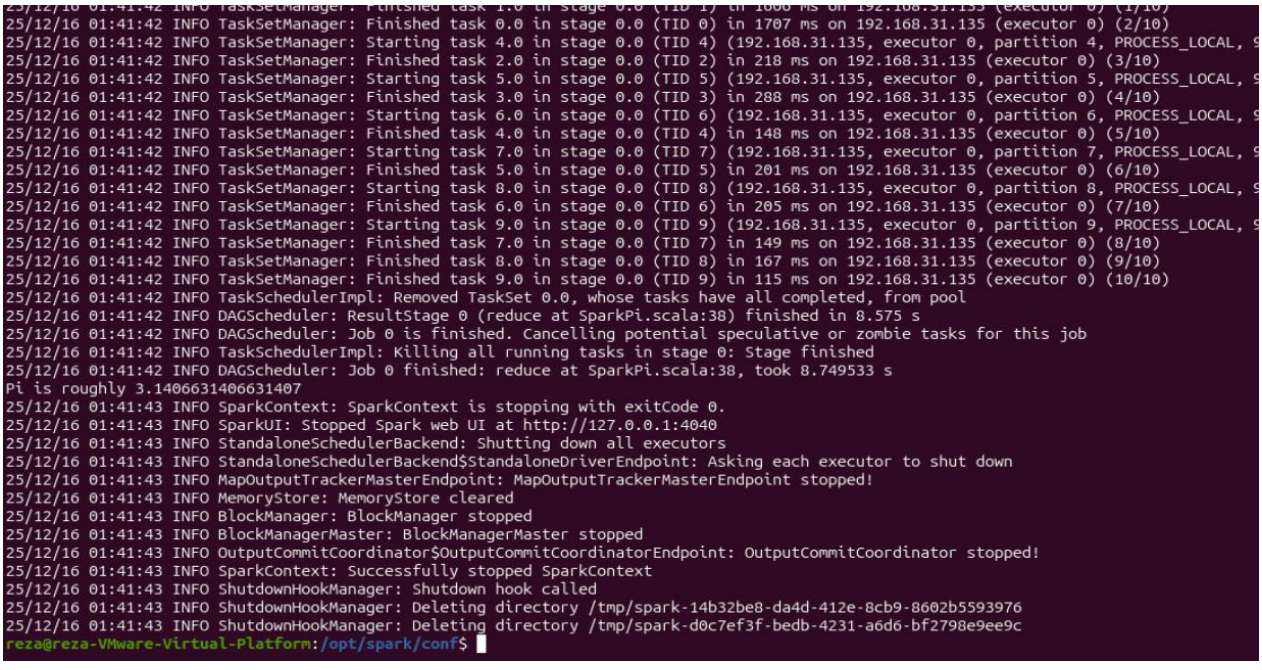


Figure 61: Implementation and output: Master/worker configuration and final integration.